Chapter 5: Functions

Python Programming Notes | Beginner Study Chapter

How functions work

A function is a reusable block of code that performs a task. Define a function using def, followed by its name, parentheses and a colon. Indent its body. The body runs when the function is called, not simply when it is defined. Functions reduce repetition and help organize a program.

Parameters and arguments

Parameters are names in a function definition. Arguments are the actual values supplied when calling it. In def add(a, b), a and b are parameters. In add(2, 3), 2 and 3 are arguments. A default parameter value is used when the caller does not supply that argument.

Returning values

return sends a result back to the caller and ends the function call. Store the result in a variable or use it in another expression. A function without an explicit return value returns None. print() displays a value, while return makes a result available to the caller.

Local variables

A variable assigned inside a function is normally local to that function. It is best to pass needed values as arguments and return results. Built-in functions such as len(), sum() and print() are already available in Python.

Try this code

```
def add(a, b):
    return a + b

result = add(2, 3)
print(result)

def greet(name="Student"):
    return "Hello, " + name

print(greet())
```